

Final Summary

The objective of this project was to analyze the probability of students passing a competitive mathematics examination using probability concepts and statistical techniques. Based on the analysis of 200 student records, several factors were identified that significantly influence the probability of passing the final examination.

1. Study Hours

Students who studied more than 10 hours per week formed an important group in the dataset. The probability of a student studying more than 10 hours was calculated using:

$$P(A) = \frac{\text{Students with Study Hours} > 10}{\text{Total Students}}$$

A higher number of study hours generally indicates better preparation, which increases the likelihood of success in the examination.

2. Attendance

Attendance was found to be one of the strongest indicators of academic performance.

The probability of having attendance greater than 80% was calculated as:

$$P(B) = \frac{\text{Students with Attendance} > 80\%}{\text{Total Students}}$$

Using Bayes Theorem, the probability that a student passed given that they had high attendance was calculated as:

$$P(\text{Pass} | \text{Attendance}) = \frac{P(\text{Attendance} | \text{Pass}) \times P(\text{Pass})}{P(\text{Attendance})}$$

The result obtained was:

$$P(\text{Pass} | \text{Attendance}) = 68.83\%$$

This shows that students with high attendance have a significantly higher probability of passing the examination.

3. Group Discussion Participation

The contingency table analysis showed that students who participated in group discussions achieved better results than the overall student population.

The conditional probability of passing given participation in group discussions was calculated using:

$$\begin{bmatrix} P(\text{Pass} \cap \text{Discussion}) \\ P(\text{Discussion}) \end{bmatrix}$$

The result obtained was:

$$\begin{bmatrix} P(\text{Pass} | \text{Discussion}) = 69.6\% \end{bmatrix}$$

Since this value is greater than the overall pass probability:

$$\begin{bmatrix} P(\text{Pass}) = 59\% \end{bmatrix}$$

it can be concluded that participation in group discussions positively influences examination performance.

4. Joint Probability Analysis

The probability that a student both participated in group discussions and passed the examination was calculated using:

$$\begin{bmatrix} \frac{\text{Students who Participated and Passed}}{\text{Total Students}} \end{bmatrix}$$

This probability demonstrated a strong relationship between collaborative learning and academic success.

5. Event Relationship Analysis

To determine whether group discussion participation and passing the examination were independent events, the following comparison was made:

$$\begin{bmatrix} P(\text{Pass}) = 0.59 \\ P(\text{Pass} | \text{Discussion}) = 0.696 \end{bmatrix}$$

Since:

$$\begin{aligned} & [\\ & P(\text{Pass} | \text{Discussion}) \neq P(\text{Pass}) \\ &] \end{aligned}$$

the events are **dependent events**, meaning participation in group discussions affects the probability of passing.

Overall Conclusion

Based on the complete probability analysis, the factors affecting the probability of passing can be ranked as follows:

1. **High Attendance (>80%)** – Strongest predictor of success.
2. **Participation in Group Discussions** – Significantly increases the probability of passing.
3. **Higher Study Hours (>10 hours/week)** – Contributes positively to academic performance.
4. **Previous Academic Performance** – Students with better previous test scores generally demonstrate higher chances of passing.

The analysis concludes that students who maintain high attendance, actively participate in group discussions, and dedicate more time to studying have the highest probability of successfully passing the competitive mathematics examination.